

Prob 1. (8 pts) Encode 2.75 using UQ3.5 and write the result in hexadecimal.

Prob 2. (16 pts total, 8 pts each) Decode the following Q3.4 code back to real numbers: (a) $C_1 = 0x77$ and (b) $C_2 = 0xAA$

Prob 3. (10 pts total) Decode Q2.5 code 0b1001_1001 to real number.

Prob 4. (32 pts total, 8 pts each) Encode the following real numbers into Q3.4 format and write the results in hexadecimal: (a) $f_1 = 0.5$, (b) $f_2 = 6.73$, (c) $f_3 = -2.25$, and (d) $f_4 = -4.5$.

Prob 5. (44 pts) Assume we use Q15 to express real numbers $f_1 = 0.5$, $f_2 = 0.25$, $f_3 = 0.125$, and $f_4 = 0.0625$. Perform the following operation in Q15: $f_A = f_1 \times f_2 + f_3 \times f_4$ where the multiplications have higher precedence than the addition. Note you need to perform the following steps:

1. (16 pts) Encode the real numbers in Q15 format to obtain I_1 to I_4 .
2. (16 pts) Perform the two multiplications using 16bit x 16bit $\Rightarrow$ 32bit format and shift your results as needed to obtain the two products in Q15 code I_{p1} and I_{p2} . (Like I_C in Figure 5 in Lctr 13.)
3. (6 pts) Add the two products I_{p1} and I_{p2} to obtain I_A , the Q15 code of f_A .
4. (6 pts) Decode I_A to obtain f_A .